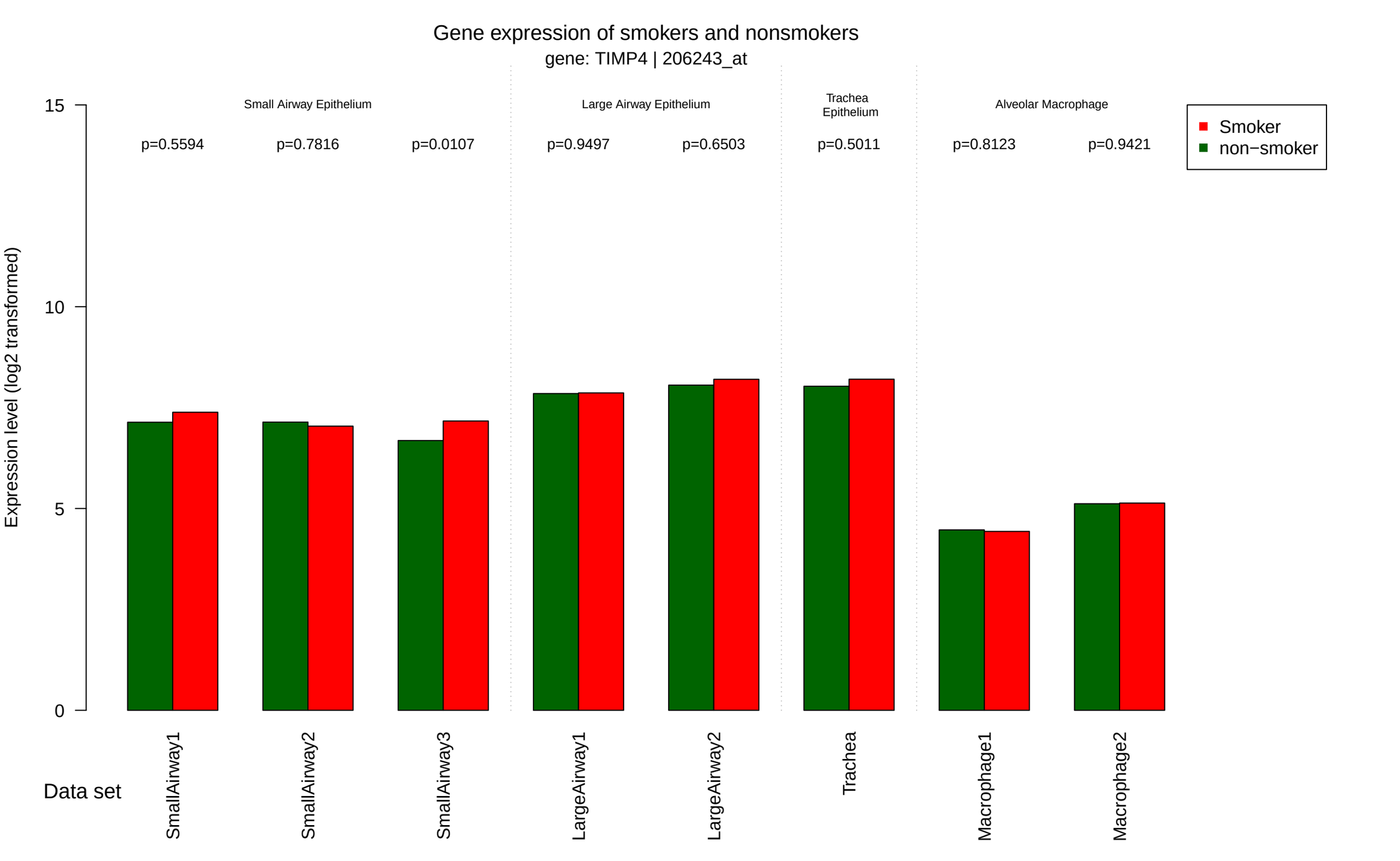
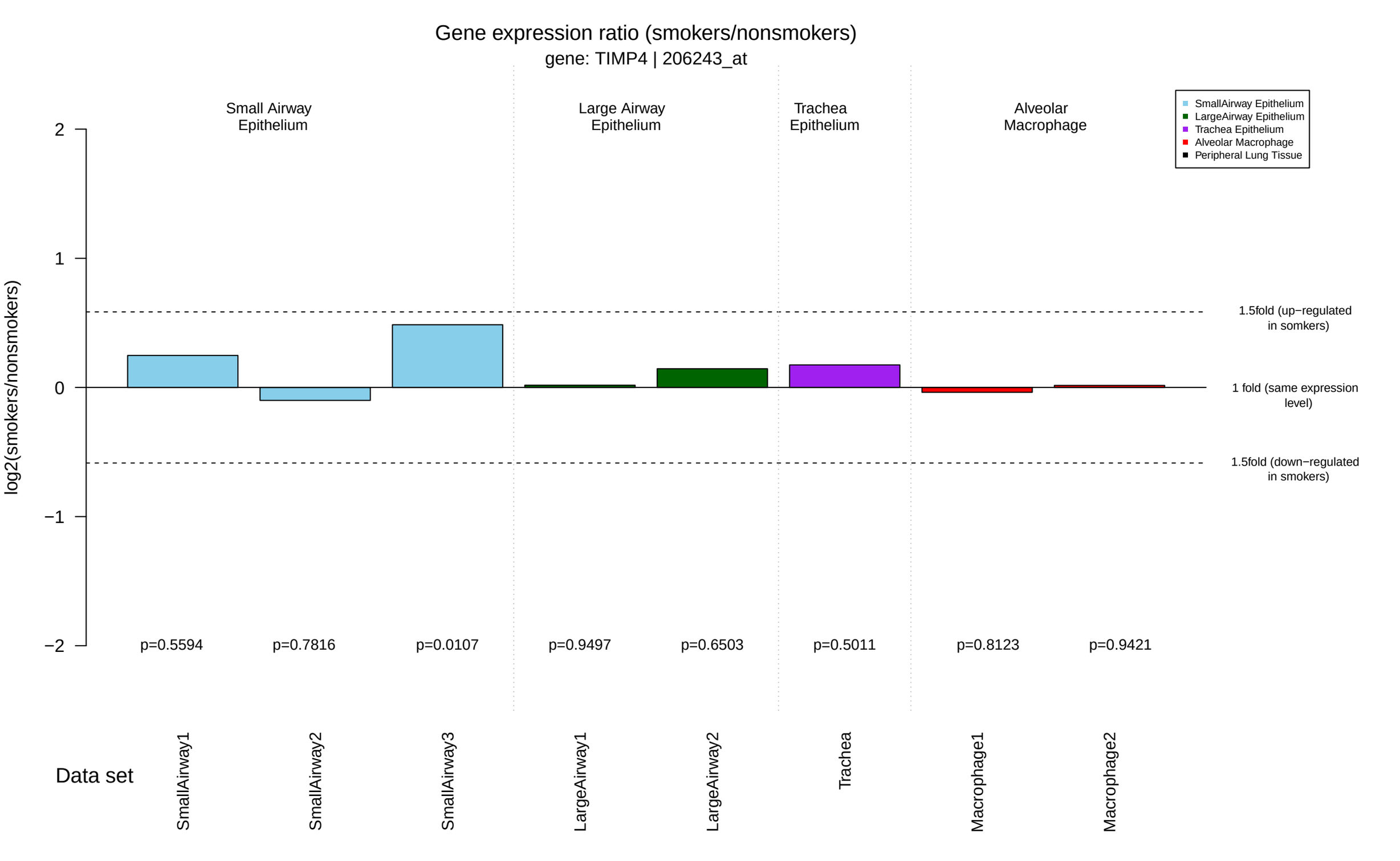
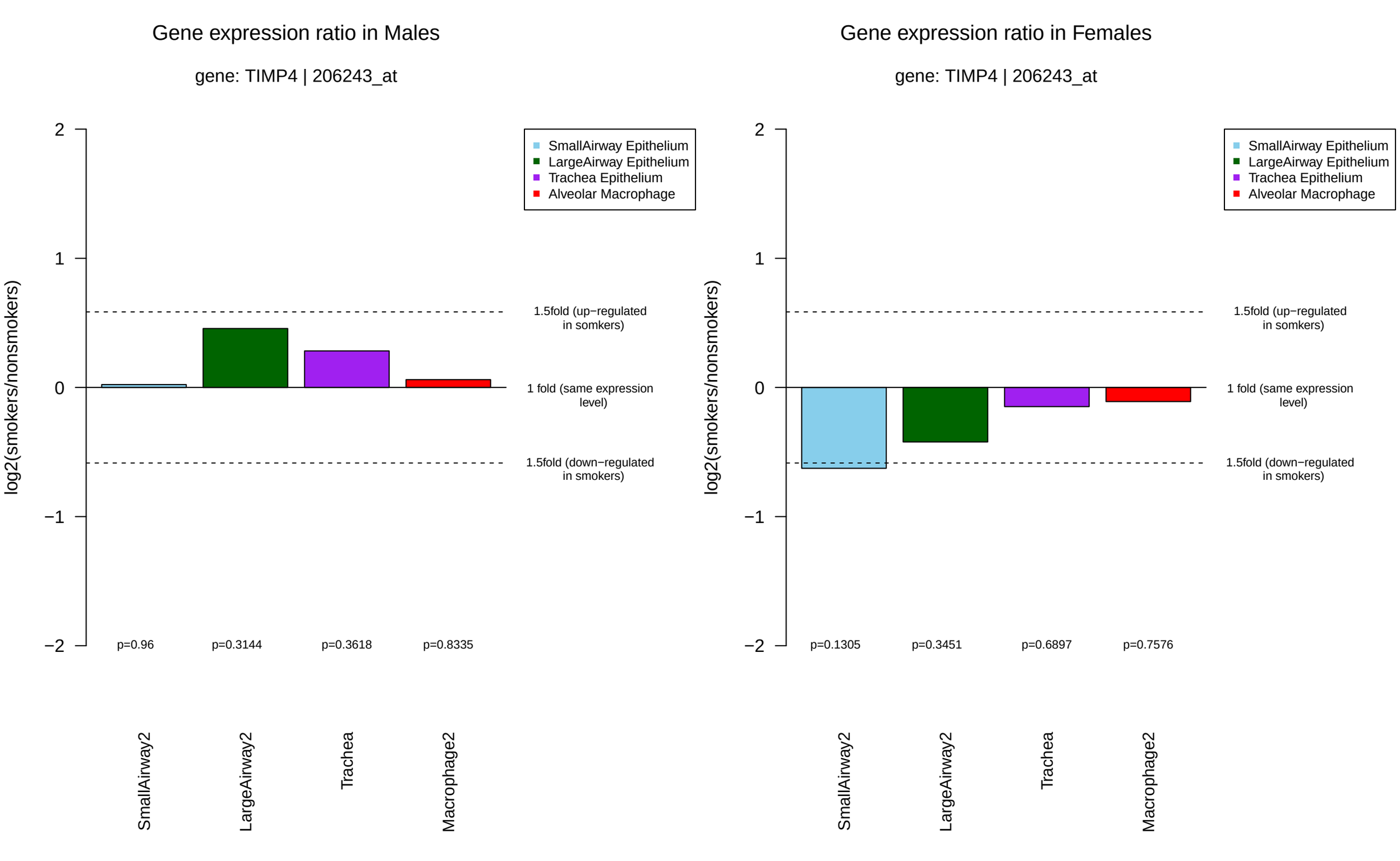
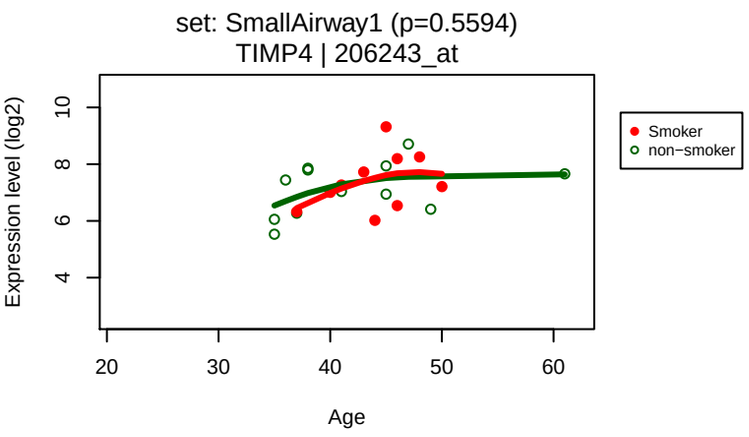
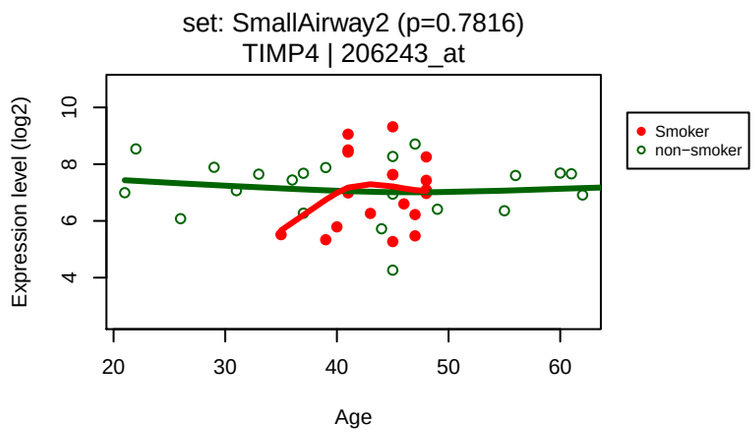
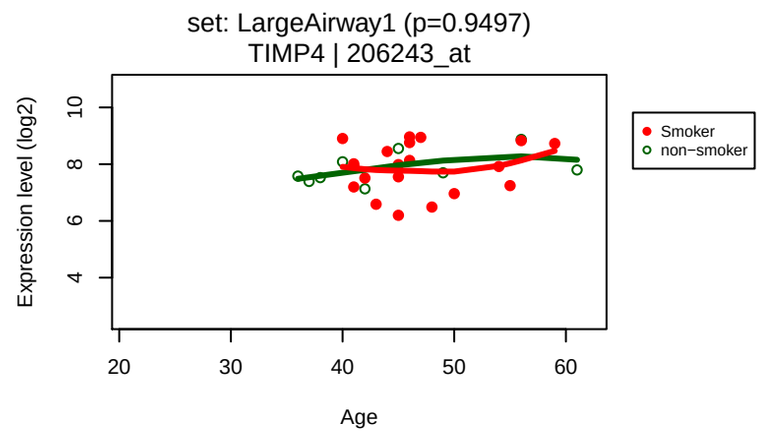

Gene expression of smokers and nonsmokers
gene: TIMP4 | 206243_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5594
p=0.7816
p=0.0107
p=0.9497
p=0.6503
p=0.5011
p=0.8123
p=0.9421
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

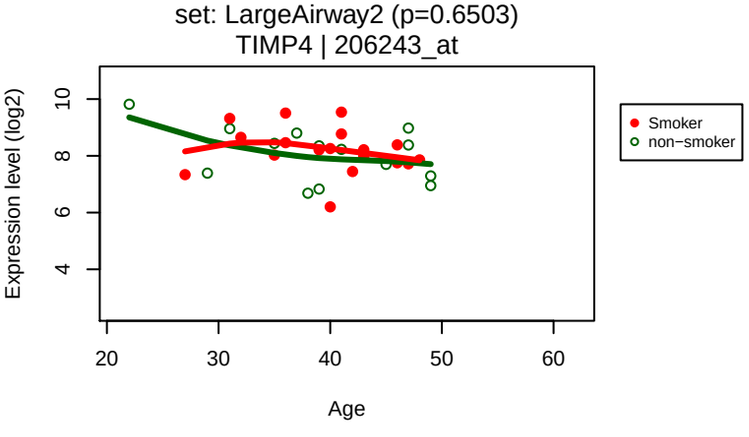

Gene expression ratio (smokers/nonsmokers)
gene: TIMP4 | 206243_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5594
p=0.7816
p=0.0107
p=0.9497
p=0.6503
p=0.5011
p=0.8123
p=0.9421
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

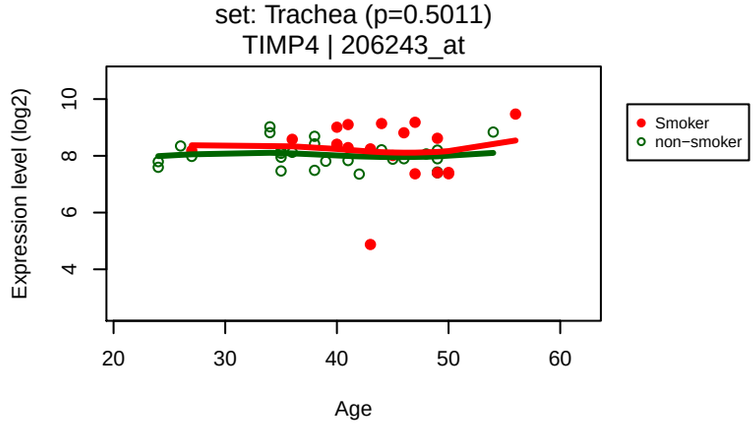

Gene expression ratio in Males
gene: TIMP4 | 206243_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.96
p=0.3144
p=0.3618
p=0.8335
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TIMP4 | 206243_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1305
p=0.3451
p=0.6897
p=0.7576
SmallAirway2
LargeAirway2
Trachea
Macrophage2

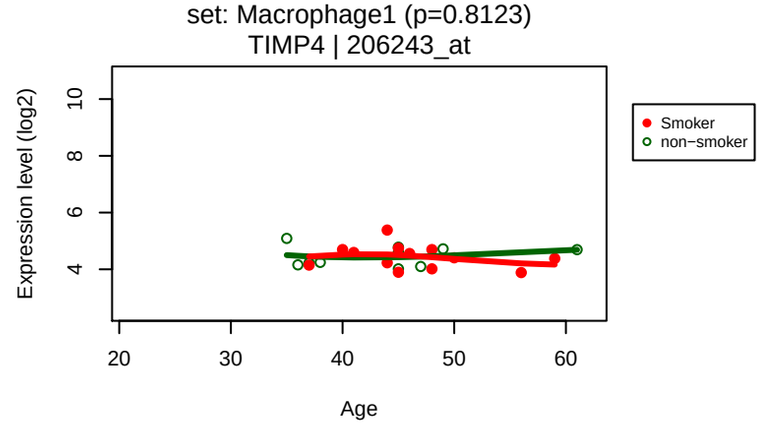

set: SmallAirway1 (p=0.5594)
TIMP4 | 206243_at
Expression level (log2)
Age
Smoker
non−smoker

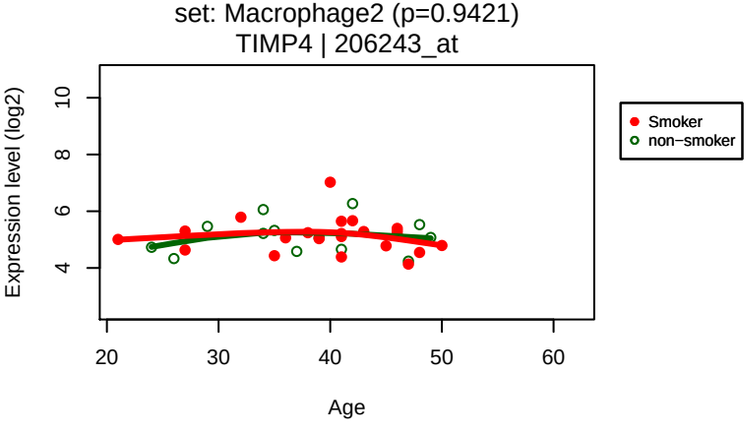

set: SmallAirway2 (p=0.7816)
TIMP4 | 206243_at
Expression level (log2)
Age
Smoker
non−smoker

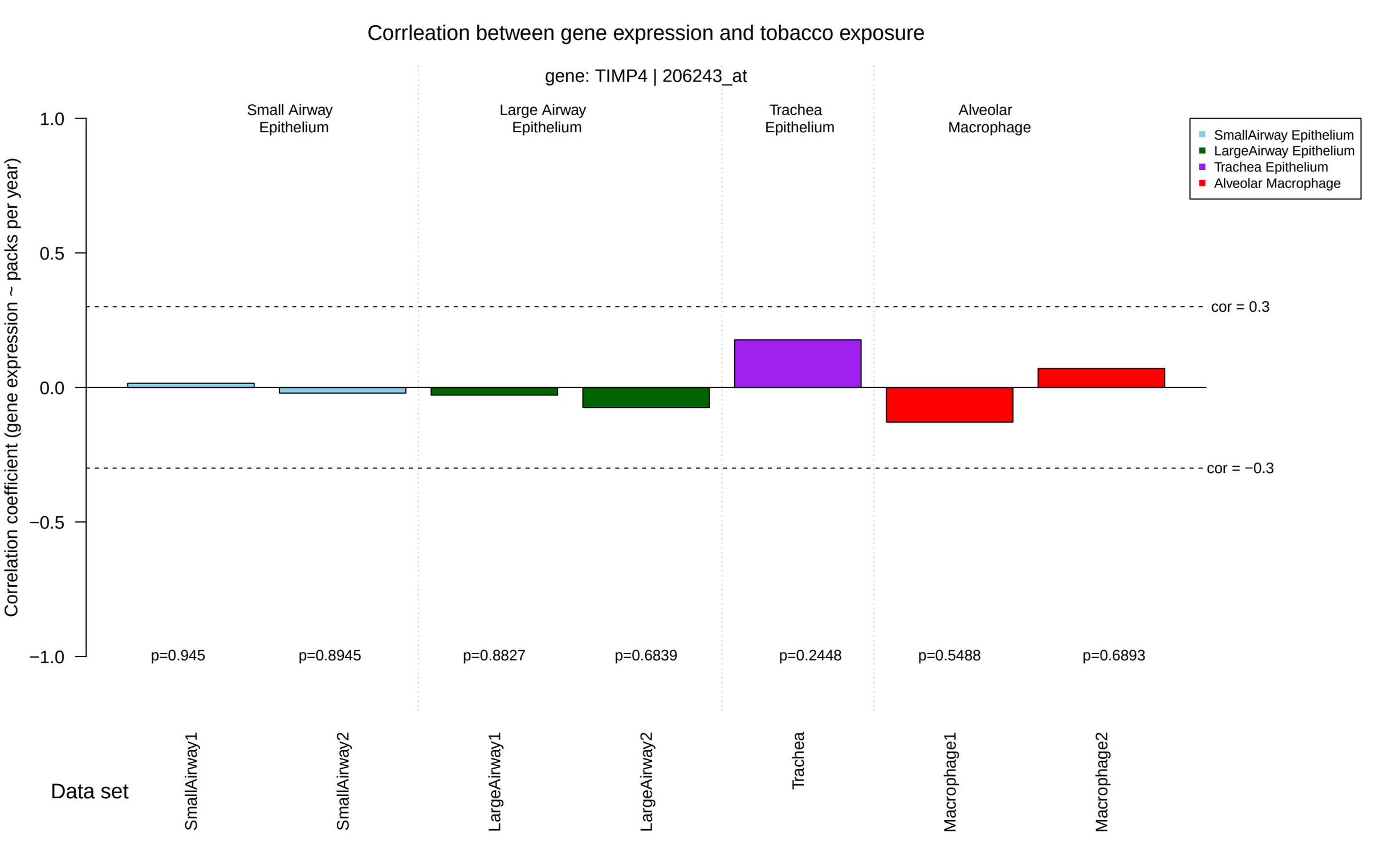

set: LargeAirway1 (p=0.9497)
TIMP4 | 206243_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6503)
TIMP4 | 206243_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5011)
TIMP4 | 206243_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8123)
TIMP4 | 206243_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9421)
TIMP4 | 206243_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TIMP4 | 206243_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.945
p=0.8945
p=0.8827
p=0.6839
p=0.2448
p=0.5488
p=0.6893
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2